Assignment 1: Markdown Buddy Use ChatGPT to Draft and Format R Docs

Reflection Answers:

1. How did AI assist in creating and improving the R project documentation?
AI was useful as a drafting and organization assistant for the synthetic retail sales analysis project. It helped propose a logical README structure that moves from the project overview to requirements, installation, usage, outputs, limitations, and attribution. This structure is important because a README should help another student or collaborator understand what the project does and how to reproduce its workflow without relying on verbal instructions. AI also suggested headings and code-chunk formatting for the R Markdown document, making the distinction between the script's purpose, inputs, processing steps, and outputs clearer.

The main benefit was the ability to refine a first draft through multiple prompts rather than writing every section from scratch. For example, an initial request for README sections could be followed by a request for concise Markdown headings and then a critique of readability and syntax. However, AI suggestions were treated as drafts rather than evidence that the documentation was correct. The repository file paths, dependencies, and example commands still need to be compared with the actual project files before submission.

2. What challenges and limitations did you identify when reviewing AI-generated Markdown?
A major limitation is that fluent documentation can appear complete even when it contains inaccurate technical details. An AI-generated README might refer to a package that is not installed, describe an output file that the script does not create, or provide a command with the wrong relative path. Markdown can also contain headings, lists, and fenced code blocks that look reasonable in plain text but render poorly when previewed. These issues matter because readers depend on the documentation to reproduce the project.

For this assignment, the proposed verification process is to compare each instruction with the actual R script, check that the dataset is synthetic, and inspect the rendered README in GitHub. The R Markdown file should be knitted in RStudio or Posit Cloud to confirm that the R chunks execute and that any tables and figures appear as intended. Numeric results should be independently checked against the underlying synthetic records rather than accepted because AI described them. A final manual spot-check should confirm consistent headings, working relative links, and clear explanations. These are verification steps to complete, not claims that the code has already been run or that every preview has passed.

3. What did you learn about ethical AI use and how would you improve this workflow?
The most important lesson is that AI can improve the clarity and speed of technical writing, but accountability remains with the person submitting the work. Ethical use requires identifying the AI tool, documenting the prompts and substantive revisions, and distinguishing AI-assisted prose from calculations or results that have been independently verified. The assignment also restricts the project to synthetic data, which avoids exposing personal or confidential information in prompts or repository files.

A stronger future workflow would begin with a working R script and a short manually written description of its inputs and outputs. AI could then propose a README outline and suggest formatting, after which each section would be checked against the actual code. Next, I would render the README, knit the R Markdown report, cross-check a sample of results, and revise unclear wording. Finally, I would keep a concise prompt appendix and an honest record of what was verified. This approach uses AI as a documentation assistant while preserving reproducibility, transparency, and personal responsibility for the final submission.

AI Assistance Declaration: I used ChatGPT (GPT-6, October 30, 2026) to draft and organize these reflection responses. The prompts concerned three reflections on AI-assisted README and R Markdown documentation, validation, and ethical use. I will verify the final text against the project files, GitHub preview, and R Markdown knit results. All final calculations must be performed and checked by me. I am responsible for the accuracy and originality of the submission.